Monitoring biodiversity for human, animal, plant and environmental health

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Abstract: The One Health approach promotes collaboration across disciplines to enhance the health of humans, animals, plants, and the environment. The Quadripartite organizations, which include the Food and Agriculture Organization of the United Nations (FAO), the United Nations Environment Programme (UNEP), the World Organisation for Animal Health (WOAH), and the World Health Organization (WHO), developed the One Health Joint Plan of Action (OH JPA) to support countries in achieving One Health. This plan consists of six action tracks, each consisting of a set of actions for implementing One Health. By requiring knowledge on zoonotic diseases (tracks 2 and 3), food and agriculture (track 4), antimicrobial resistance (track 5), and environmental health (track 6), most of these tracks directly concern biodiversity. However, there are currently no indicators for monitoring the OH JPA. Our research examines the extent to which all six tracks are covered by the Kunming-Montreal Global Biodiversity Framework (KM-GBF) of the Convention on Biological Diversity (CBD), which contains many indicators at the intersection of biodiversity and health. We assessed (1) the link between each indicator of the KM-GBF and human, animal, plant, and environmental health and (2) the usability of these indicators for monitoring One Health actions. We found that 75% of indicators are associated with health, and that a similar proportion can be used for monitoring One Health actions. Overall, our work aims to strengthen collaboration between the CBD Secretariat and the Quadripartite Organizations by highlighting the need for shared data, policy, and governance practices.

Keywords: biodiversity indicators, Kunming-Montreal Global Biodiversity Framework, One Health, One Health Joint Plan of Action, Quadripartite Organizations

Introduction

- The One Health approach
  - Interconnection between human, animal, plant, and environmental health
  - Zoonotic diseases, non-communicable diseases, food safety, antimicrobial and antiparasitic resistance, climate change, pollution
  - Collaboration across disciplines
- The One Health Joint Plan of Action
  - Quadripartite Organizations
    - World Health Organization (WHO)
    - World Organisation for Animal Health (WOAH)
    - Food and Agriculture Organization of the United Nations (FAO)
    - United Nations Environment Programme (UNEP)
  - 6 action tracks, many actions, even more activities
  - No indicators
- The Kunming-Montreal Global Biodiversity Framework
  - Convention on Biological Diversity
  - Protecting biodiversity by working towards targets and goals
  - Global Action Plan recognizes that biodiversity is linked with health
- Monitoring framework of the KM-GBF
  - Types of indicators (headline, binary, component, complementary)
  - Many indicators are linked with health (examples)
  - Reusing indicators decreases the workload of countries
- Objectives of our study
  1. Assess the link between biodiversity indicators and human, animal, plant, and environmental health
     - Strengthens the link between biodiversity and health
     - Reinforces the need for collaboration across disciplines
     - Highlights the need for shared policy and governance practices between the CBD Secretariat and the Quadripartite Organizations
  2. Evaluate the usability of indicators for monitoring One Health actions
     - Highlights the need for data sharing between Parties, organizations, and other stakeholders

- Reduces the workload on countries

Methods

We qualitatively assessed the relevance of all 204 KM-GBF indicators (Conference of the Parties to the Convention on Biological Diversity, 2025) to Health. Specifically, we evaluated the degree to which they were linked to each of the four pillars of One Health (i.e., to human, animal, plant, and environmental heath), as well as their usability for monitoring the OH JPA. Our assessments were informed by the metadata (rationale, definition, method of computation, data sources, and scale of use) of the indicators provided by the World Conservation Monitoring Centre of the UN Environment Programme (UNEP-WCMC, 2025), when available, looking for explicit considerations of Health. When such metadata was unavailable, assessments were entirely based on expert judgment. The result of this analysis and the code to reproduce the figures are available on Zenodo.

Assessing the link between indicators and Health

Assessing the link between the KM-GBF indicators and the four pillars of One Health requires a working definition of human, animal, plant, and environmental health. We defined each of these groups of species based on the main objects of interest of the Quadripartite organizations. Humans are their own pillar and the primary focus of the WHO. We considered pets, livestock, and edible marine and freshwater species captured from fisheries and aquaculture in the animal health pillar. These are species mainly looked after by veterinarians and food inspectors and that are of core concern to the WOAH and FAO. Even though humans and wildlife are also animals, we excluded them from the animal health pillar. We included all cultivated plants used for food, fuel, and medicine into the plant health pillar, i.e. plant species that are the primary focus of the FAO. Wildlife was considered in the environment health pillar, alongside all other species not included in the other three pillars. The environment health pillar also includes whole ecosystems (natural and artificial), as well as tree species used in forestry and wild fish species. It is the primary focus of the UNEP. These definitions are alined with the core contributing sciences of the One Health approach (Gibbs, 2014; Lerner & Berg, 2017).

We used a broad definition of Health that considers the overall well-being of individuals and their capacity to function normally, which includes the absence of diseases (WHO, 1948). In humans and domestic and wildlife animals, well-being is the ability for an individual to satisfy

its physical, mental, and behavioral needs (Webb et al., 2019), whereas diseases are undesirable conditions often leading to pain, suffering, or death. In plants, which are not conscious organisms (Mallatt et al., 2021), health is more ill-defined (Döring et al., 2012). Here, we considered plant health as the extent to which an individual is able to function physiologically, and plant diseases as impediments to their normal physiological functioning often leading to death. Similarly, when it comes to whole ecosystems, consisting of many individuals and their interactions, health is also ill-defined (Schaeffer et al., 1988). We defined environmental health as the extent to which an ecosystem can maintain its biological and chemical processes and adapt to changes. Disturbances are the ecosystem analogue of diseases, and represent degradations that usually lead to a decline in ecosystem functioning. When assessing environmental health, we also considered the health of species included in the environment pillar of One Health, e.g. wildlife and uncultivated plant species. These descriptions guided us in our assessments.

We classified the degree to which the indicators are linked to each of the four pillars of One Health into four categories: direct connection, indirect connection, potential connection, and no connection. An indicator was qualified as directly connected to Health when there is a direct causal relationship between the indicator and Health. This is the case when the indicator directly measures a condition or determinant of health. For example, headline indicator 7.2 "Pesticide environment concentration and/or aggregated total applied toxicity" was considered directly connected to human health because of the toxicity of many pesticides under high level of exposure (Kim et al., 2017). An indicator was qualified as indirectly connected to Health when there is an intermediary factor between the indicator and Health. For instance, component indicator 4.CT.1 "Number of plant and animal genetic resources for food and agriculture secured in either medium- or long-term conservation facilities" was considered indirectly connected to human health because conserving genetic resources increases the resilience of food systems to global disasters (Esquinas-Alcázar, 2005), which in turn protects human health. An indicator was qualified as potentially connected to Health when there are multiple intermediary factors between the indicator and Health, or when its connection to Health is likely but difficult to certify. For example, complementary indicator 2.CY.2 "Proportion of key biodiversity areas in favourable condition" was qualified as potentially connected to human health because key biodiversity areas are mainly defined with ecological criteria for conserving rare or endemic species and pristine habitats (Butchart et al., n.d.), and may or may not include areas of high importance for human health. Finally, an indicator was qualified as not connected to Health when the connection is unlikely or absent. This was the case for component indicator 21.CT.3 "Index of linguistic diversity", for which

we did not find any evidence of a link to human, animal, plant or environmental health. For each of the four pillars of One Health, we counted the number of indicators assigned to each category.

Assessing the usability of indicators for monitoring the OH JPA

We evaluated the usability of the KM-GBF indicators for monitoring the OH JPA by judging if they can likely detect trends relevant to the actions of the plan. For each action track, we classified the usability of the indicators into three categories: directly usable, usable after adaptation, and not usable. Directly usable indicators can be used in their current form to monitor at least one action in an action track. They do not require adaptations. For example, we qualified component indicator 4.CT.4 "Proportion of local breeds classified as being at risk of extinction" as being directly usable for monitoring the fourth action track because it directly measures food safety risks. In contrast, indicators usable after adaptation need to be slightly modified before they can be used to monitor actions in an action track. These adaptations should be small changes in the scale of measurement, the data resolution, or the taxa monitored by the indicator. For instance, we qualified headline indicator 9.1 "Benefits from the sustainable use of wild species" as being usable after adaptation for monitoring the second action track on zoonotic epidemics and pandemics. To be more relevant for this action track, this indicator could refer more specifically to wildlife reservoirs of zoonotic agents instead of all wild species. Finally, not usable indicators are outside the scope of an action track or need to be greatly modified before being used to monitor an action track. For example, we qualified component indicator 4.CT.3 "Human-wildlife conflict indicator" as being not usable for monitoring the fifth action track on antimicrobial resistance.

We assessed usability for each action track independently, i.e. an indicator can be useful for monitoring multiple action tracks. For each action track, we counted the total number of indicators in each category. In addition, we counted the number of usable indicators for the whole plan regardless of the action tracks. Then, we partitioned these numbers based on the indicator type in the KM-GBF Monitoring Framework (i.e. headline, binary, component, and complementary indicators) and the thematic categories of the Global Action Plan on Biodiversity and Health. Finally, for each action track where an indicator was considered usable (either directly or after adaptation), we identified the most relevant action that can be monitored by the indicator, and counted the number of indicators associated with each action.

Results and discussion

Link between biodiversity indicators and health

- Most indicators are linked with health
  - How many indicators are directly or indirectly linked with human, animal, plant, and environmental health?
  - Description of Figure 1
  - Examples of indicator linked with human, animal, plant, and environmental health
- Link between biodiversity and health
  - The state of biodiversity impacts health
  - Ecosystem services benefit health
  - Biodiversity and health have similar pressures
  - Biodiversity and health are protected with similar actions

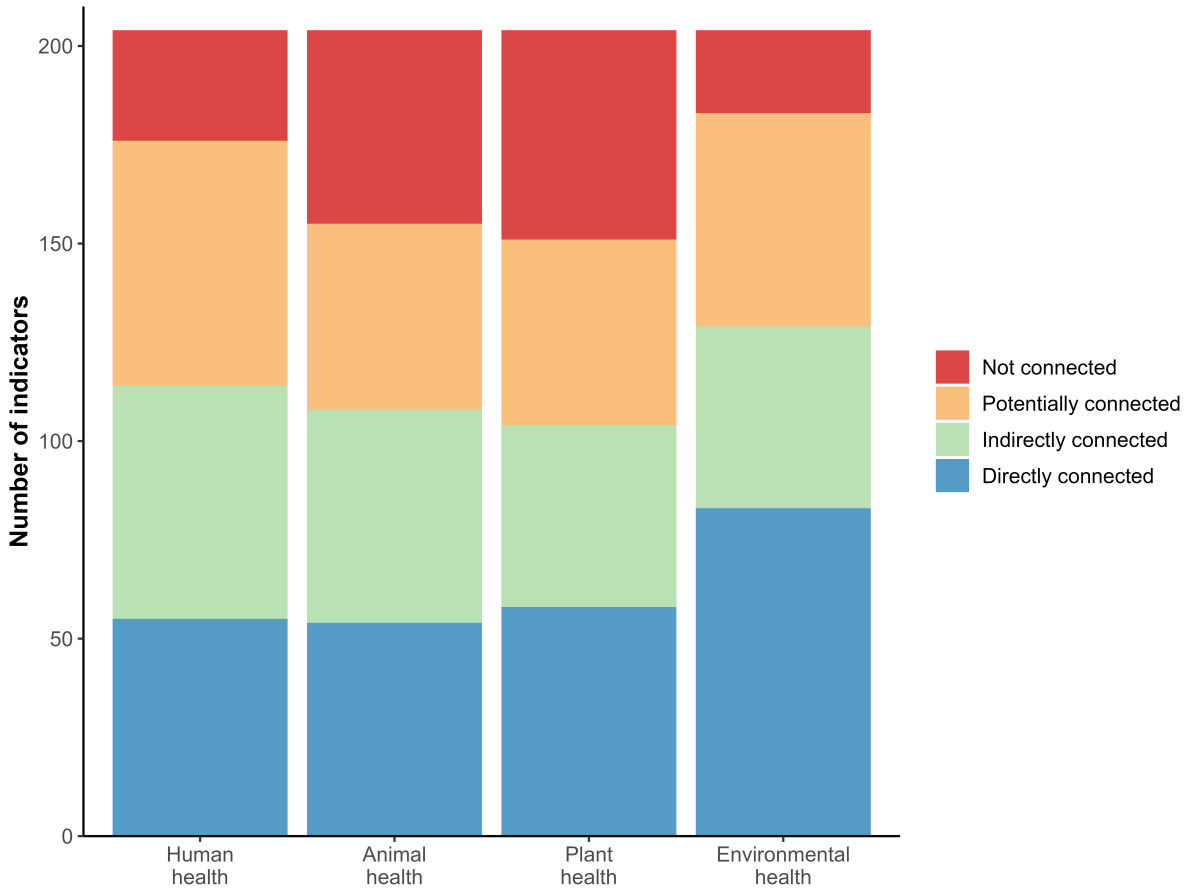

Figure 1: **Number of KM-GBF indicators linked with human, animal, plant, and environmental health.** Each indicator was classified as being either directly connected (blue), indirectly connected (green), potentially connected (orange), or not connected (red) to each of the four pillars of One Health.

Usability of indicators for monitoring the OH JPA

- Most indicators can be used to monitor the OH JPA
  - How many indicators for each action track?
  - Description of Table 1
  - Description of Figure 2
  - Importance of directly reusing indicators
  - Indicators usable after adaptation are based on similar and robust methodologies, which minimizing training requirements
- Many usable indicators are headline and binary indicators
  - How many?
  - Presentation of important gaps
  - Description of Figure 3

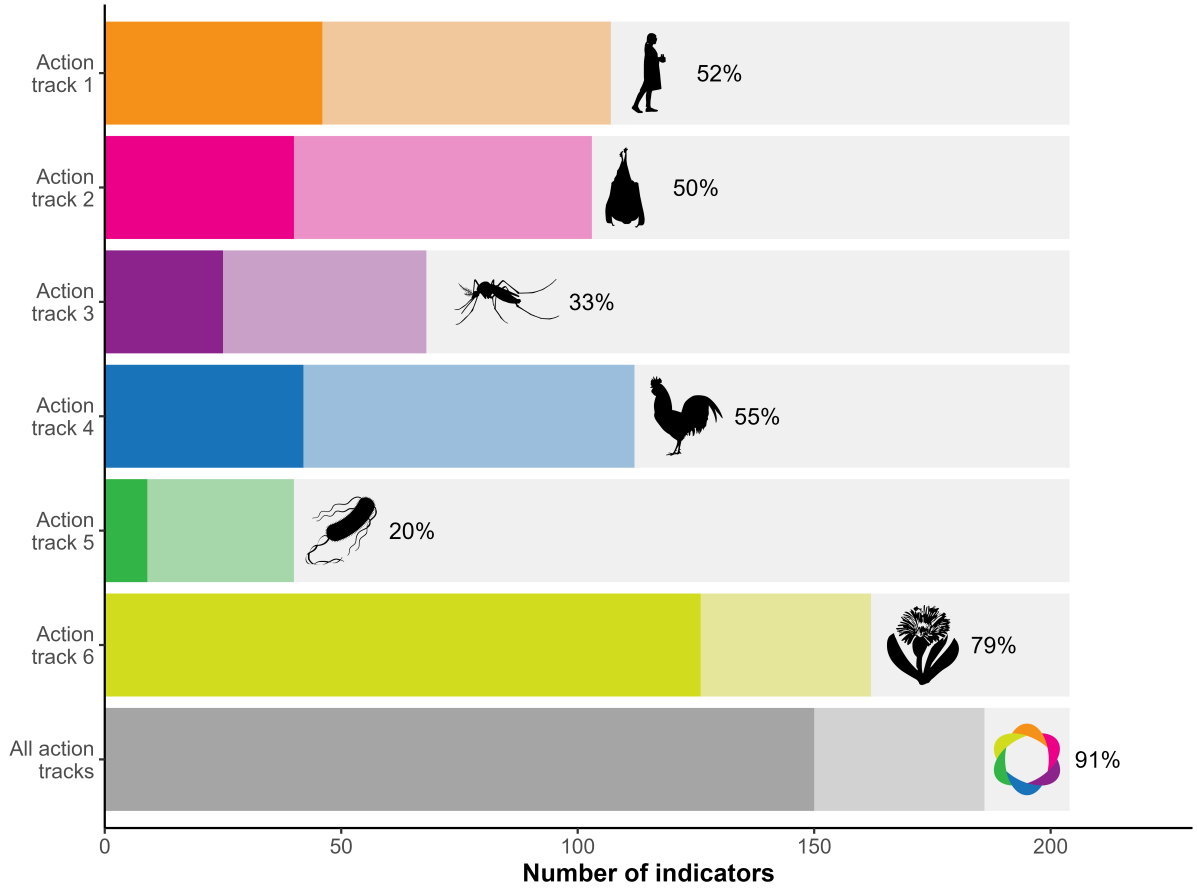

Figure 2: **Number of KM-GBF indicators that can be used to monitor at least one action in each of the OH JPA action tracks.** The opaque bars represent directly usable indicators, whereas the transparent bars represent those usable after adaptation. The proportion of usable indicators (either directly or after adaptation) for each action track is indicated. The bottom bar and proportion designate the indicators that can monitor at least one action in the whole plan.

Table 1: **Number of KM-GBF indicators that can be used directly and after adaptation to monitor each action of the OH JPA.** The total number of usable indicators for each action track is indicated in the highlighted blue lines. A specific indicator can be used to monitor multiple action tracks, but only the most relevant action in each action track was identified.

Action	Directly usable	Usable after adaptation	Total usable
1 Enhancing One Health capacities to strengthen health systems	**46**	**61**	**107**
1.1 Establish the foundations for One Health capacities	30	5	35
1.2 Generate mechanisms, tools, and capacities to establish a One Health competent workforce and the frameworks/processes to facilitate One Health work	8	50	58
1.3 Generate an enabling environment for the effective implementation of One Health	8	6	14
2 Reducing the risks from emerging and re-emerging zoonotic epidemics and pandemics	**40**	**63**	**103**
2.1 Understand the drivers of emergence, spillover, and spread of zoonotic pathogens	16	27	43
2.2 Identify and prioritize targeted, evidence-based upstream interventions to prevent the emergence, spillover, and spread of zoonotic pathogens	7	6	13
2.3 Strengthen national, regional, and global One Health surveillance, early warning, and response systems	17	30	47
3 Controlling and eliminating endemic zoonotic, neglected tropical and vector-borne diseases	**25**	**43**	**68**
3.1 Enable countries to develop and implement community-centric and risk-based solutions to endemic zoonotic, neglected tropical, and vector-borne disease control using a One Health approach involving all relevant stakeholders	3	14	17
3.2 Ensure the harmonized application of One Health principles at all levels by implementing practical measures to strengthen local, national, regional, and global policy frameworks for the control and prevention of endemic zoonotic, neglected tropical, and vector-borne diseases	18	10	28
3.3 Increase political commitment and investment in the control of endemic zoonotic, neglected tropical, and vector-borne diseases, by advocating for and demonstrating the value of a One Health approach	4	19	23
4 Strengthening the assessment, management and communication of food safety risks	**42**	**70**	**112**
4.1 Strengthen the One Health approach in national food control systems and food safety coordination	18	30	48
4.2 Utilize and improve food systems data and analysis, scientific evidence, and risk assessment in developing policy and making integrated risk management decisions	15	21	36
4.3 Foster the adoption of the One Health approach in national foodborne disease surveillance systems and research for the detection and monitoring of foodborne disease and food contamination	9	19	28
5 Curbing the silent pandemic of antimicrobial resistance (AMR)	**9**	**31**	**40**
5.1 Strengthen the capacity and knowledge of countries to prioritize and implement context-specific collaborative One Health work to control AMR in policy, legislation, and practice	7	17	24
5.2 Reinforce global and regional initiatives and programmes to influence and support One Health responses to AMR	2	13	15
5.3 Strengthen global AMR governance structures	0	1	1
6 Integrating the environment into One Health	**126**	**36**	**162**
6.1 Protect, restore, and prevent the degradation of ecosystems and the wider environment	92	16	108
6.2 Mainstream the health of the environment and ecosystems into the One Health approach	9	7	16
6.3 Integrate environmental knowledge, data, and evidence into One Health decision-making	20	6	26
6.4 Create an interoperable One Health academic and in-service training programme for environmental, medical, agricultural, and veterinary sector professionals	5	7	12

    - Important because these are mandatory indicators that are more likely to be measured
- Usable indicators are in all categories of the Action Plan

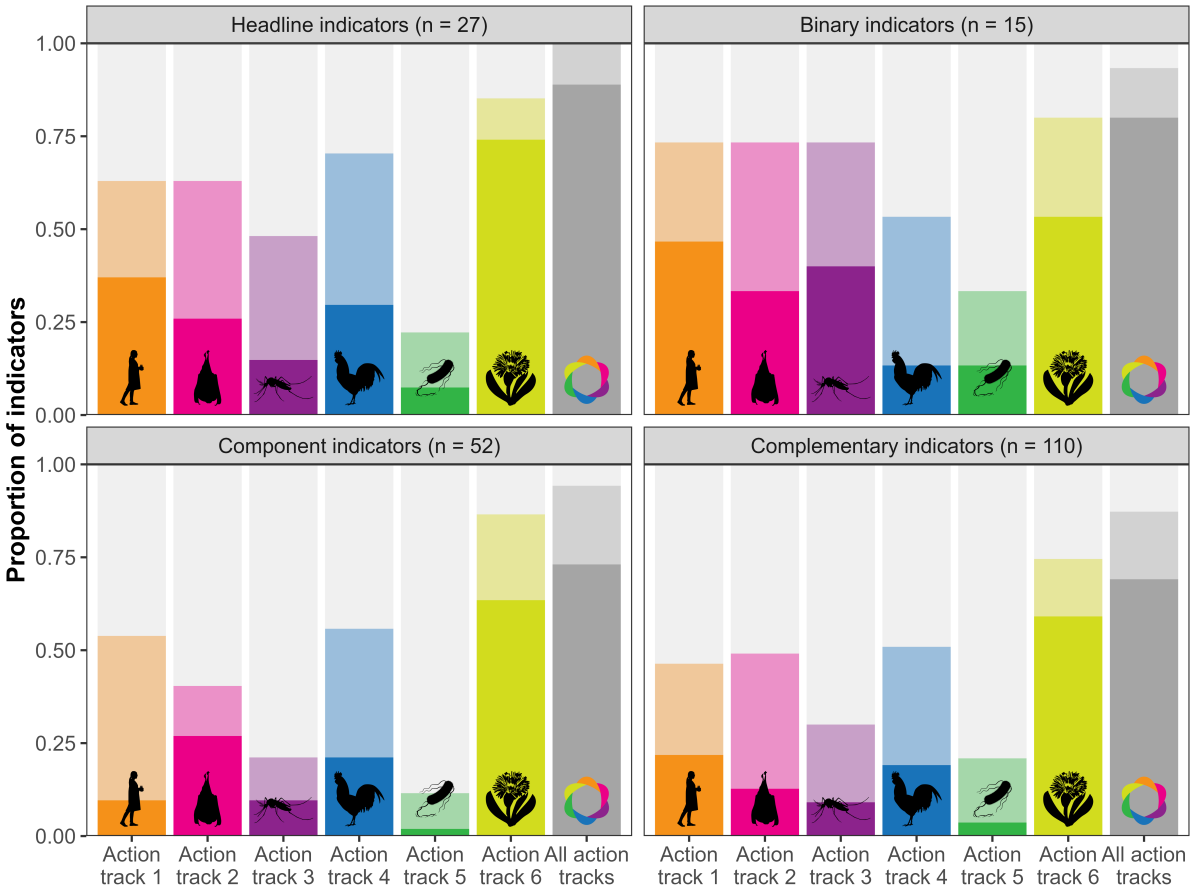

Figure 3: **Proportion of KM-GBF indicators that can be used to monitor at least one action in each of the OH JPA action tracks, partitioned by the different groups of indicators.** The number of indicators in each group is indicated in parentheses. The opaque bars represent directly usable indicators, whereas the transparent bars represent those usable after adaptation. The gray bars designate the indicators that can monitor at least one action in the whole plan.

- The KM-GBF addresses many dimensions of health
- Presentation of important gaps
- Description of important categories and their connection with health

Conclusion

- Monitoring Frameworks
  - The monitoring framework of the OH JPA can be based on indicators of the KM-GBF
  - Importance of reusing indicators to decrease workload on countries
  - Importance of sharing data, methodologies, and expertise
  - Sharing policy and governance practices
- Gaps in indicators
  - Indicators

Figure 4: **Proportion of KM-GBF indicators that can be used to monitor at least one action in each of the OH JPA action tracks, partitioned by the different thematic categories of the Global Action Plan on Biodiversity and Health.** The number of indicators whose target is in each thematic category is indicated in parentheses. The opaque bars represent directly usable indicators, whereas the transparent bars represent those usable after adaptation. The gray bars designate the indicators that can monitor at least one action in the whole plan. Indicators that monitor KM-GBF goals A, B, C, and D are not assigned to any thematic category.

- Other indicators could be identified in other monitoring frameworks (e.g. SDG)
  - New indicators can be developed after identifying gaps

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